

OPDI Papers

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Methodology studies behind the [Open Performance Data Initiative](#).

Each paper benchmarks an OPDI algorithm against EUROCONTROL reference data and reports what the comparison actually showed, including where a proposed method failed. Every figure renders from a committed CSV cache — no credentials, no database, no cluster — so the results on any page can be reproduced from the repository alone.

1 Papers

[Detecting departure and destination aerodromes from ADS-B — version 4.1](#) · [PDF](#)

Moves detection into the OPDI pipeline itself: H3 cells as the index, exact great-circle distance as the comparison, three modes behind one parameter. Finds that the detection radius barely matters for coverage while the height threshold decides almost everything — but that read on accuracy the two axes pull the other way, so a coverage-only view overstates the case for a high threshold. The production baseline improves enough under the new candidate stage to overturn version 3's conclusion. Explains the preprocessing pipeline, each mode, and every metric from first principles.

[Detecting departure and destination aerodromes from ADS-B — version 3](#) · [PDF](#)

Finds that six of the seven most frequent departure errors were a scheduled civil airport losing to an adjacent military airfield, and removes that class entirely with a scheduled-service preference. Fixes an out-of-area rule that was pre-empting valid aerodrome matches, reports out-of-area recall and precision separately for the first time, and carries a production patch proposal for `flights.py`.

[Detecting departure and destination aerodromes from ADS-B — version 2](#) · [PDF](#)

Corrects two accounting errors in version 1. A large share of flights have an origin or destination outside the ingested area, and scoring those as detection failures understated every method; and the aircraft filter was an allowlist that discarded 47.74% of the fleet to exclude the 8.64% that are rotorcraft. Adds trajectory extrapolation (which fails), the cleaning ablation, the aerodrome-set sweep, an error breakdown, and a measurement of `track_id` quality.

[Detecting departure and destination aerodromes from ADS-B — version 1](#) · [PDF](#)

A comparison of seven ADEP/ADES methodologies against APDF ground truth over two three-day samples a year apart. Finds that the production algorithm’s altitude-trend logic contributes no accuracy over a nearest-aerodrome rule on the same flights, and that replacing it with explicit endpoint abstention improves departure detection by 8–10 points of coverage — while the same change fails to replicate for arrivals, and is not claimed.